A Holding-Constraint Roadmap for the Two-Arm Bar Carry

A technical report on the PRM pilot (`holding_prm`) and its first measurements

husky_assembly_tamp `motion_planner` — written for Yijiang Huang

30 September 2026; second version 1 October 2026

Abstract

The task-space BiRRT that plans the M1 two-arm bar carry (`plan_constrained_dual_arm`; see `docs/cdfm_task_space_birrt_report.tex`) holds the bar exactly at every waypoint and loses most of its extensions to the 10-degree joint-continuity gate. This report describes the opposite planner, proposed in the meeting of 29 September 2026 and built as a pilot in `motion_planner/holding_prm/`: a probabilistic roadmap whose nodes hold the bar exactly, whose edges are straight lines in joint space (so the joints never jump), and whose edge cost is how far the two tool flanges drift from the bar along the line. The query searches from a *set* of start configurations to a *set* of goal configurations, so the start pose and both branch choices are chosen by the search rather than fixed beforehand. The pseudocode is written from the code; lines that differ from the BiRRT are printed in red and tagged [P-n], collected in Section 9. Section 11 reports the first measurements on bar B18, goal 2 (the goal whose 807 starts the BiRRT was run on in full) and on three goals of bar B3.

Second version (1 October 2026). A refinement turns roadmap routes into paths that hold the bar exactly (Section 6): on B18 goal 2 it gives an exact path for 34 of the 36 starts within 10 mm and for 245 starts in all, every one of which passes the production repair at 2 mm steps, against 21 starts the BiRRT solved. New experiments: wider edges (E1), every IK branch at equal draws (E4), the refinement (E2, E3), a search for better carry poses (E5) and overnight precomputation (E6). Terms are defined in *Words used* (Section 1).

Contents

1 What is being planned, and what changes **2**

2 Where the code lives **5**

3 Building blocks **6**
- 3.1 The holding violation . . . 6
- 3.2 Straight lines in joint space, and why continuity is free . . . 6
- 3.3 Turns . . . 6
- 3.4 Collision . . . 7

4 Building the roadmap **7**
- 4.1 Anchors . . . 7
- 4.2 Sampling: forward kinematics for one arm, IK for the other . . . 7
- 4.3 Edges . . . 8

5 Querying the roadmap **9**
- 5.1 Endpoints . . . 9
- 5.2 The gap of every node . . . 9
- 5.3 Lazy collision checking . . . 10
- 5.4 The questions asked . . . 10

6 Refinement: from a roadmap route to a path that holds the bar exactly **11**
6.1 Tracking one edge exactly . . . 11
6.2 Exact gaps: tracking routes until they all track . . . 13
6.3 Tube draws around a route (way 2) . . . 13
6.4 A BiRRT seeded with the route (way 1) . . . 13
6.5 The production check . . . 13

7 Running it in parallel **14**

8 Storage and the dashboard **14**

9 Core differences from the task-space BiRRT **15**

10 Parameters and their defaults **16**

11 What the pilot measured **16**
11.1 What was run . . . 16
11.2 How fast a joint-space line leaves the holding surface . . . 16
11.3 The roadmaps and what they cost . . . 17
11.4 Starts 0 and 12, and all 807 starts, against the node budget . . . 19
11.5 Which starts are near the goal: classes, carry anchors, and the BiRRT . . . 20
11.6 The best path . . . 21
11.7 Seeds and two sampler choices . . . 22
11.8 From roadmap routes to exact paths (E2) . . . 24
11.9 Tightening routes: tube draws (way 2) and a seeded BiRRT (way 1) (E3) . . . 26
11.10 Wider edges (E1) and every IK branch at the same number of draws (E4) . . . 27
11.11 Re-calibrating the three carry poses (E5) . . . 28
11.12 Precomputing overnight (E6) . . . 28
11.13 Bar B3 . . . 30
11.14 What this says, and what it does not . . . 30

12 Reading guide **32**

1 What is being planned, and what changes

The movement is the same as in the BiRRT report: both UR5e arms of the Husky hold one bar rigidly and carry it from a carry (home) pose to the approach pose in front of the structure; the base does not move. The notation is the same too: $q = (q_L, q_R) \in \mathbb{R}^{12}$ are the twelve arm joints; $G_L, G_R \in \mathrm{SE}(3)$ are the grasps (the pose of each `tool0` flange in the bar frame); FK_L, FK_R are the flange forward kinematics. A configuration holds the bar when both flanges agree on one bar pose,

$$\mathrm{FK}_L(q_L)\, G_L^{-1} \;=\; \mathrm{FK}_R(q_R)\, G_R^{-1}, \tag{1}$$

and the set $\mathcal{M}$ of such configurations is a six-dimensional surface in the twelve-dimensional joint space.

The two ways to treat the two constraints. A path has to satisfy two things at once: every configuration holds the bar (Eq. 1), and consecutive configurations are close in joint space (no joint moves more than $\tau = 10°$ between waypoints). The BiRRT makes the first one exact and tests the second; its night-1 benchmark measured that 75 % of its extensions stop on the second. The roadmap swaps the roles:

	task-space BiRRT	holding roadmap (this report)
holding the bar	exact at every waypoint (IK)	exact at nodes; *measured* along edges
joint continuity	tested per waypoint (the gate)	true by construction (straight joint lines)
what a failure means	the tree stops growing	the edge costs more
answer	a path or nothing	the smallest holding error any path needs

The last row is the point of the pilot. The BiRRT cannot tell "no path exists" from "the planner did not find it". The roadmap always returns a number: the smallest tolerance on the holding error at which some start connects to some goal. A small number means near-feasible; a large one means the start and goal branch sheets are probably not connected.

Only the goal pose is given. The export fixes the goal bar pose T_g and nothing else: neither the start pose nor either configuration. The BiRRT must pick one start configuration and one goal configuration before it searches (its difference D-9), and a chain of heuristics does that (goal branch pairing with the home anchors, backward tracking). The roadmap does not need to choose:

- the **goal set** is every collision-free analytical branch pair at T_g (up to 8×8 left-right combinations), each on the library's turn and on the home turn (Section 3.3);
- the **start set** is every collision-free branch pair at every candidate start pose; the pilot takes the benchmark library's surveyed carry poses (807 for B18 goal 2): the three carry anchors (horizontal, vertical, on the back), each shifted on a grid of $4 \times 4 \times 4$ offsets (± 0.1 and ± 0.3 m per axis) and rolled about the bar in twelve 30° steps, keeping the poses that can be held without collision (the library has no yaw variants);
- one search runs from a virtual *source*, joined at zero cost to every start node, to a virtual *sink*, joined at zero cost to every goal node. The cheapest path therefore chooses the start pose, the start branch pair and the goal branch pair together, and the report says which it chose.

Because the grasps are fixed per bar, $\mathcal{M}$ is the same surface for every goal of a bar: one roadmap per bar and scene can serve all of its goals, and a query only adds its own start and goal nodes.

Words used in this report

word	meaning
branch pair	One of the (up to 8×8) ways two arms can hold the bar at one pose: each arm's analytical IK has up to eight solutions (shoulder left/right, elbow up/down, wrist flipped or not), and a branch pair is one left solution with one right solution. Different branch pairs are physically different robot postures.
turn	The same posture written with one or more joints shifted by 360°. Shoulder pan (±720°), shoulder lift and the three wrists (±240–270°) can each reach one angle in two (pan: three) ways. Physically identical; different to a planner that measures raw joint differences.
anchor	A configuration the sampler centres its draws on; anchors are also roadmap nodes. They are the home configuration, the goal's branch pairs and a spread of library start configurations, each on up to two turns (Section 3.3).
node budget	The number of *sampled* nodes a build keeps (the `-nodes` flag). Anchors come on top.
β (max branches)	How many of the second arm's IK solutions are kept per draw. $\beta = 1$ keeps only the solution nearest the draw's centre (the default).
difficulty class	The benchmark library's label for a start, from a straight-line probe from the goal to the start: *free* (the straight bar move works and is collision-free), *blocked* (it works but collides on the way), *broken* (the IK breaks or jumps on the way).
carry anchor	Which of the three carry poses a start was derived from: horizontal (bar along y in front of the robot), vertical (bar along z in front) or back (bar along x over the base).
gap (of a node)	The smallest holding tolerance at which this node reaches some goal node through collision-free roadmap edges: the worst edge score on its best route (Section 5). A property of the whole route, not of the node's own position.
per-start gap	For one start pose, the smallest gap over all its branch pairs and turns.
component	A set of nodes joined to each other by collision-free edges; two components have no edge between them. "Largest" is the node count of the biggest one.
Collides(q)	The production full-cell collision check (robot, tools, attached bar, environment); true when q is in collision.
VerifyGaps	The loop that collision-checks only the edges on each start's best route, searches again, and repeats until those routes are fully checked (Alg. 8).
exact gap	(Section 6) The smallest roadmap tolerance at which a start's route can be turned into a path that holds the bar exactly.

2 Where the code lives

Function	File	Role
holding_violation, violation_from_tools	holding.py	the violation measure (Alg. 1)
tool0_world, build_holding_model	holding.py	numpy forward kinematics on the cached arm bases
interpolate_confs, rebranch12	holding.py	joint-space lines; moving a configuration onto a 2π turn
draw_driver, lift_draw, sample_chunk	sampling.py	the node sampler (Algs. 2, 3)
arm_branches	sampling.py	one arm's analytical IK on the cached base
expand_bar_pose	sampling.py	every collision-free branch pair at a bar pose (Alg. 6)
make_collision_fn	sampling.py	the production full-cell collision check
candidate_pairs, evaluate_edge	roadmap.py	nearest neighbours; measuring one edge (Alg. 4)
edge_cost, Roadmap	roadmap.py	the penalised weight; the node and edge tables
minimax_to_goals, route_from	search.py	every node's gap to the goal set (Alg. 7)
build_graph, shortest_between_sets	search.py	source/sink search (Dijkstra)
lazy_search, path_report	search.py	lazy collision checking (Alg. 8); path profiles
WorkerPool, init_worker, task_*	parallel.py	worker processes (Section 7)
build_roadmap, query_roadmap	pilot.py	the two drivers (Algs. 5, 9)
save_roadmap, copy_dashboard	store.py	files under <root>/prm/; the dashboard page
track_edge, fine_walk, track_edge_landing	tracking.py	the exact tracker (Section 6)
Tracker, refine_query, pretrack_roadmap	refine.py	exact gaps, assembly, tube rounds; overnight tracking
carry_bar_pose, score_carry_pose	carry.py	carry poses and their scores (Section 11.11)
plan_seeded_birrt	dual_arm_task_space_rrt/seeded.py	the BiRRT seeded with a route (imports the core)
main	scripts/prm_pilot.py	command line: build, query, refine, track, export, info
validate, birrt	scripts/prm_bridge.py	the production check; the seeded BiRRT (moves the core's log first)
stage1, starts, stage2, pick	scripts/prm_carry_search.py	the carry search
page	benchmark/dashboard/prm.html, prm-viewer.js	the roadmap viewer (Section 8)
tests	benchmark/tests/test_holding_prm.py	23 checks, including one that the package never imports the planner core

Paths without a folder are in husky_assembly_tamp/motion_planner/holding_prm/. What is reused from the rest of the code base: ssik_inprocess.solve and ssik_inprocess.fk (the generated per-arm solvers and their forward kinematics), ssik_ik.rebranch_toward_seed, ARM_SLICE and ARM_BASE_LINK, and from the benchmark load_trimmed_problem, open_scene, state_with_conf12 and provenance.

Kept apart from the BiRRT work on purpose. Other sessions were editing the planner core, the benchmark runner and the benchmark dashboard while this was written. The pilot therefore lives in new files only, writes only under <root>/prm/ (never runs/ or the batch catalogue runs.json), has its own command line, and never imports dual_arm_task_space_rrt/core.py: that module opens its log file in "w" mode when imported, which empties a log a running batch may be writing. Three places where the benchmark reaches the core indirectly are avoided by writing short local versions: the collision check (SceneCtx.collision_fn imports the core for its explain helper), the library loader (benchmark.library imports benchmark.sampling, which imports the core) and the static-site copy. A test runs the import in a fresh process and asserts no core module is loaded; the log file's size and time stamp were unchanged across every pilot run.

3 Building blocks

3.1 The holding violation

Algorithm 1: HoldingViolation — `holding_violation` / `violation_from_tools`

function HoldingViolation(q)
  $F_L \leftarrow W_L\,\mathrm{fk}_L(q_L)$; $F_R \leftarrow W_R\,\mathrm{fk}_R(q_R)$ ▷ W: arm base in the world, read once per scene
  $F_R^\star \leftarrow F_L\,G_L^{-1}G_R$ ▷ where the bar placed by the left flange wants the right one
  $F_L^\star \leftarrow F_R\,G_R^{-1}G_L$ ▷ and the other way round
  $e_p \leftarrow \max(\|p(F_R^\star) - p(F_R)\|\,,\ \|p(F_L^\star) - p(F_L)\|)$
  $e_r \leftarrow \max(\angle(F_R^\star, F_R),\ \angle(F_L^\star, F_L))$ ▷ angle by atan2(sin, cos)
  return (e_p, e_r)
end function

Three details. (i) fk is the ssik artifact's own forward kinematics, generated from the calibrated URDF like the solver; it takes about 30 μs and agrees with PyBullet to 3.4×10^{-7} on every transform entry (measured on 45 library configurations of B3). So the measure needs no simulator, and the parent process can score whole paths. (ii) Taking the larger of the two directions makes the measure independent of which arm is the reference: the two angles are equal, the two distances differ by the lever arm between the grasps. (iii) The angle of a nearly identity rotation is taken from both its sine and cosine; an arc-cosine alone loses about half the digits, which on the dashboard showed as a spurious 0.1° error at nodes that hold the bar exactly. On every library configuration tested (primary goals, goal branch pairs, starts) the measure returns 10^{-7} m and 2×10^{-7} rad, four hundred times below the production FK gate.

One number for a tolerance. A tolerance is one number t in millimetres. A configuration passes it when $e_p \le t\,\mathrm{mm}$ *and* $e_r \le \rho\,t$ degrees, with $\rho = 0.4°/\mathrm{mm}$ by default (the "5 mm and 2 degrees" pair of the pilot plan). Equivalently its *score*

$$s(q) = \max(1000\,e_p,\ e_r^\circ/\rho) \quad [\mathrm{mm}] \tag{2}$$

is at most t. Edges store e_p and e_r separately, so ρ and t can be changed without re-measuring anything.

3.2 Straight lines in joint space, and why continuity is free

Interp$(q_a, q_b; \delta)$ (`interpolate_confs`) returns the points of the straight joint-space line cut into $\lceil \|q_b - q_a\|_\infty / \delta \rceil$ equal pieces; the difference is taken on raw joint values, the same rule as the BiRRT's gate (`core.joint_step_deltas`). With $\delta \le \tau$, every piece of every edge would pass the production continuity gate, so no path the roadmap returns can contain a branch flip. The tests check the bound; every path report re-measures the largest step between its dense points.

3.3 Turns

Four of the six UR joint ranges are wider than one turn (shoulder pan $\pm2\pi$, shoulder lift ±4.71, wrists ±4.19), so one physical posture can be written as two or three joint vectors 2π apart. A roadmap that measures raw joint differences sees these "twins" a full turn apart, and an edge between them would be a full-turn swing. Rebranch(q, r) (`rebranch12`, which applies `ssik_ik.rebranch_toward_seed` per arm) moves every joint onto the turn nearest a reference r when that stays in limits. The pilot uses two references: the configuration *as given* (the library's own turn) and the *home* configuration.

Which nodes get more than one copy. A sampled node is stored once: on the turn nearest the centre of the draw that made it. Only anchors and query endpoints (start and goal branch pairs) are stored on both turns, and only when the two copies differ; the query reports which copy a path used. So the roadmap does not hold every 2π copy of every node; it holds at most two copies of the few configurations the search has to start or end at.

This matters in practice: for B18 goal 2, start 0's right shoulder pan sits at $-319°$ in the library, while the goal's branch pairs sit at $+50°$ or $+204°$; its home-turn twin is $+41°$, nine degrees from three of them.

3.4 Collision

The check is the production one: copy the template cell state, write the twelve joints, and let compas_fab test self-collision, the tools, the attached bar and the environment with the scene's allowed-collision lists (`make_collision_fn`; the same call as `api._build_cfab_collision_fn`). It costs about 15 ms and dominates everything else the pilot does: one ssik solve is 1.8 ms, one forward-kinematics pair 60 μs. A 2π twin is the same physical pose, so it is never checked twice.

4 Building the roadmap

4.1 Anchors

Draws are centred on *anchors* (`make_anchors`): the home configuration, every collision-free branch pair recorded for the chosen goals (`free_branch_pairs`), and a spread of library starts per goal (40 by default, picked across the three difficulty classes and the three carry anchors as the benchmark's `select_starts` does), each on its given and its home turn. Anchors are also inserted as nodes. For B18 goal 2 there are 81: the home configuration, the goal's 7 branch pairs, and 73 from the 40 anchor starts (40 as given, plus 33 whose home-turn copy differs; the home-turn copies of home and of the goal pairs coincide with them). A global box spanned by all anchors is available but off: the 81 anchors of B18 goal 2 span 214° to 580° per joint (the right shoulder pan the most), i.e. the full range, and draws from that box are the wound-up arms the pilot plan warned about.

4.2 Sampling: forward kinematics for one arm, IK for the other

Algorithm 2: DrawDriver — `draw_driver`

function DrawDriver(mode, driver)
  $a \leftarrow$ uniform anchor index; $c \leftarrow$ anchor$[a]$ ▷ c: the 12-joint centre of the draw
  if mode = `anchor_box` **then** $q_d \sim \mathcal{U}(c_d \pm m)$ ▷ $m = 90°$ per joint
  else if mode = `anchor_gauss` **then** $q_d \sim \mathcal{N}(c_d, \sigma^2)$ ▷ $\sigma = 30°$
  else if mode = `anchor_line` **then**
    $b \leftarrow$ an anchor of a different role (start ↔ goal) ▷ a start–goal line when possible
    $c \leftarrow c + u\,(\text{anchor}[b] - c)$, $u \sim \mathcal{U}(0,1)$; $q_d \sim \mathcal{N}(c_d, \sigma_\ell^2)$ ▷ $\sigma_\ell = 10°$
  end if
  return (clip(q_d, limits), c, a)
end function

Algorithm 3: SampleChunk — `sample_chunk` / `lift_draw`

function SampleChunk(seed, k, n)
  rng $\leftarrow$ `SeedSequence`(seed, k) ▷ chunk k gives the same nodes on any worker

for $i = 1 \dots n$ **do**
driver $\leftarrow$ L if $(kn + i)$ even else R; other $\leftarrow$ the other arm ▷ both arms drive equally often
mode $\sim$ mix; $(q_d, c, a) \leftarrow$ DrawDriver(mode, driver) ▷ default mix 0.2 / 0.3 / 0.5
$T \leftarrow \mathrm{FK}_{\mathrm{driver}}(q_d)\, G_{\mathrm{driver}}^{-1}$ ▷ the bar pose this arm implies [P-1]
$B \leftarrow \mathrm{IK}_{\mathrm{other}}(T\, G_{\mathrm{other}};\ \mathrm{seed} = c_{\mathrm{other}},\ \mathrm{rescue\ off})$ ▷ re-branched to c, nearest first [P-8]
$B \leftarrow B[1{:}\beta]$ ▷ $\beta = 1$: the branch nearest the centre
for $q_o \in B$ **do**
$q \leftarrow (q_d, q_o)$; **assert** HoldingViolation$(q) \le 10^{-4}$ ▷ a failure stops the build
if not Collides(q) **then** keep q with (mode, driver, a, geometry)
end if
end for
end for
end function

Each draw gives the second arm up to eight IK solutions; β is how many of them become nodes, nearest the draw's centre first. Keeping only the nearest ($\beta = 1$) was a measured choice. On a small smoke roadmap (bar B3) with a node budget of 300 sampled nodes, keeping every solution split the roadmap into 103 components: the second arm's median distance to the anchor was 182°, i.e. most kept nodes were flipped elbows or wrists that sit far from every other node and join nothing. With $\beta = 1$, the same budget of 300 nodes gave 44 components. The B18 comparison at 2000 nodes is in Table 6, and Section 11.10 repeats it with the same number of draws instead of the same number of nodes. Other branch sheets are still represented, because every branch pair of the goal is an anchor and draws around an anchor are seeded from that anchor.

4.3 Edges

Algorithm 4: EvaluateEdge — `evaluate_edge`

function EvaluateEdge(q_a, q_b)
$(e_p^{\max}, e_r^{\max}, t^\star) \leftarrow$ max of HoldingViolation over Interp$(q_a, q_b; \delta_v)$ ▷ $\delta_v = 2.5°$, nearly free [P-3]
$P \leftarrow$ interior points of Interp$(q_a, q_b; \delta_c)$ ▷ $\delta_c = 10° = \tau$ [P-2]
for j **in** bisection order of P **do** ▷ middle first, then quarters, ...
if Collides(P_j) **then return** (free = false, $t_{\mathrm{collide}} = j/|P|$, $e_p^{\max}$, $e_r^{\max}$)
end if
end for
return (free = true, $e_p^{\max}$, $e_r^{\max}$, $t^\star$, $\|q_b - q_a\|_2$, $\|q_b - q_a\|_\infty$)
end function

Candidate pairs (`candidate_pairs`) are each node's $k = 10$ nearest neighbours in the largest-joint-move metric (a `cKDTree` with $p = \infty$) no further than 90°, symmetrised and sorted so the work splits into the same chunks every time. The violation is measured four times more finely than collision because it costs two forward-kinematics calls per point; collision is checked at the continuity step, which is also what the production gate would look at. The end nodes are already known to be free.

Algorithm 5: BuildRoadmap — `build_roadmap`

function BuildRoadmap(bar, goals, N)
start the worker pool (one scene per process) ▷ Section 7
anchors $\leftarrow$ `make_anchors`(goals); insert the collision-free ones as nodes

4: **while** fewer than N sampled nodes kept **do**
5: issue chunks $k, k{+}1, \ldots$ of SAMPLECHUNK in parallel ▷ round size from the measured yield
6: **end while**
7: keep the first N sampled nodes in chunk order ▷ same nodes for any worker count
8: pairs ← `candidate_pairs`(nodes, k, radius)
9: edges ← EVALUATEEDGE on every pair, in parallel (collision skipped with `-lazy`)
10: save `roadmap.json.gz` and `config.json`; add the run to `prm/index.json`
11: **end function**

5 Querying the roadmap

5.1 Endpoints

Algorithm 6: EXPANDBARPOSE — `expand_bar_pose`

1: **function** EXPANDBARPOSE(T, references)
2: $B_L \leftarrow \mathrm{IK}_L(TG_L)$; $B_R \leftarrow \mathrm{IK}_R(TG_R)$ ▷ all branches, rescue on (one-off solve)
3: **for** $(q_L, q_R) \in B_L \times B_R$ **do** ▷ up to 64 pairs [P-5]
4: **if** COLLIDES(q_L, q_R) **then**
5: **continue**
6: **end if**
7: **for** (name, r) **in** references **do** ▷ given: the library configuration; home
8: $q' \leftarrow$ REBRANCH($(q_L, q_R), r$); merge with an equal q' already listed, adding the name
9: **end for**
10: **end for**
11: **return** every distinct q' with its pair index and turn names
12: **end function**

The goal pose and every chosen start pose are expanded in parallel, one pose per task. Each start node carries five labels, all different things (see *Words used*): its start index (which library start pose), carry anchor (horizontal, vertical or back), difficulty class (free, blocked or broken: the library's straight-line probe), branch-pair index (which of the up to 64 postures at that pose) and turn (which 2π copy of that posture); plus whether it is the library's own configuration. The expansion must find the library's `free_branch_pairs` again (they are the same pairs, solved by the same solver); a test checks this on the smoke goal, every query records the count, and every B18 query found all 7 of goal 2's pairs.

5.2 The gap of every node

Algorithm 7: MINIMAXTOGOALS — `minimax_to_goals`

1: **function** MINIMAXTOGOALS($\mathcal{G}$, goal set)
2: weight every usable edge by its score s (Eq. 2); join a sink to every goal node at 0
3: $M \leftarrow$ minimum spanning tree of $\mathcal{G}$ (Kruskal)
4: gap(sink) ← 0
5: **for** each tree edge (u, v) in breadth-first order from the sink **do**
6: gap(v) ← max(gap(u), s_{uv}); parent(v) ← u
7: **end for**

8: **return** gap, parent ▷ gap = ∞ for nodes not joined to any goal
9: **end function**

In a minimum spanning tree, the tree path between two nodes has the smallest possible largest edge of any path between them. With the sink joined to the goal set, one tree gives *every* node its *gap*: the smallest tolerance at which a chain of free edges joins it to some goal node, and the next node on that chain. The best start is the start node with the smallest gap; a start index's gap is the smallest gap over its branch pairs and turns. On the dashboard any clicked node's best route to the goals is simply followed through the parent pointers.

5.3 Lazy collision checking

Expanding every start of a goal gives thousands of endpoint nodes and tens of thousands of candidate endpoint edges; checking all of them for collision would cost more than the roadmap. The query measures the holding error of every endpoint edge (cheap) but checks collision only where it decides an answer. VerifyGaps does this for the gaps: it first computes every node's gap as if all unchecked edges were free, collision-checks only the edges on each start's best route, drops the ones that collide, and repeats until every start's best route consists of checked, free edges.

Algorithm 8: VerifyGaps and LazySearch — `query_roadmap` / `lazy_search`

1: **function** VerifyGaps
2: **repeat**
3: (gap, parent) ← MinimaxToGoals with unchecked edges assumed free [P-7]
4: U ← unchecked edges on the route (via parent) of each start index's best node
5: collision-check U in parallel; mark the colliding ones blocked
6: **until** $U = \varnothing$
7: **end function**

8: **function** LazySearch(find_path)
9: **repeat**
10: P ← find_path(edges free or unchecked) ▷ filtered or penalised Dijkstra, source → sink
11: U ← unchecked edges of P; check them in parallel; mark the colliding ones
12: **until** P = None **or** $U = \varnothing$
13: **return** P
14: **end function**

Both loops are exact, not approximations. Assuming unchecked edges free can only lower a gap or shorten a path; when the deciding route comes back with every edge checked and free, nothing cheaper exists in the true graph, since the true graph is a subset of the optimistic one. For the best node of each start index this makes the gap exact; the gaps of other nodes stay optimistic lower bounds and are only used for colouring.

5.4 The questions asked

Algorithm 9: QueryRoadmap — `query_roadmap`

1: **function** QueryRoadmap(roadmap, goal, starts, tolerances)
2: goal nodes ← ExpandBarPose(T_g, {given: primary, home}) [P-5]
3: start nodes ← $\bigcup_s$ ExpandBarPose(T_s, {given: q_s, home}); reuse roadmap nodes at the same q

endpoint edges ← EvaluateEdge without collision on each new node's k nearest (radius 90°)
VerifyGaps() ▷ per-start gap, exact
best ← start node of smallest gap; report its route (the minimax route)
report the shortest path within the best gap (LazySearch, filtered at gap)
for t **in** tolerances $(2, 5, 10, 30\,\text{mm})$ **do** [P-6]
filtered: LazySearch over edges with $s \le t$, weight $= \|\Delta q\|_2$
penalised: LazySearch over all free edges, weight $= \|\Delta q\|_2 + w \max(0, s/t - 1)$
starts connected ← #{start indices with gap $\le t$}
end for
save the query: endpoint tables, final edge verdicts, every node's gap and parent, the paths with their dense profiles
end function

Each path report stores the node list, the start and goal it used (start index, carry anchor, branch pair, turn; goal pair, turn), the dense joint path at δ_v, the holding error at every dense point, where the worst point sits, and the largest joint step between dense points (at most δ_v by construction). The penalised weight uses $w = 10$: going one tolerance over costs as much as ten radians of joint travel.

6 Refinement: from a roadmap route to a path that holds the bar exactly

A roadmap route within a tolerance t is a chain of straight joint-space edges whose holding error stays under t; its nodes hold the bar exactly, the points between them do not. The refinement turns such routes into paths that hold the bar *exactly*, at every waypoint, without giving up continuity. It is built in four layers, each tried only where the one before it fails (code: `holding_prm/tracking.py`, `refine.py`; `scripts/prm_bridge.py` and `dual_arm_task_space_rrt/seeded.py` for the parts that use the planner core).

6.1 Tracking one edge exactly

Algorithm 10: TrackEdge — `tracking.track_edge`

---

function TrackEdge(q_a, q_b; seed s (default q_a))
knots $k_0 = q_a, \ldots, k_m = q_b$ on the joint line, every 2.5° of largest joint move
$T_i \leftarrow$ the bar pose halfway between the two arms' bar poses at k_i ▷ exact at q_a, q_b
waypoints: $T_i \to T_{i+1}$ cut into $\lceil \max(d/\Delta_p,\ \theta/\Delta_r) \rceil$ equal pieces ▷ the BiRRT's rule, $\Delta_p = 1\,\text{cm}$, $\Delta_r = 0.025\,\text{rad}$
$p \leftarrow s$
for each waypoint bar pose T, with $g =$ the joint line's configuration at the same place **do**
for each arm **do**
$B \leftarrow \text{IK}(TG)$, every branch moved onto p's turn ▷ rescue off
$B' \leftarrow \{q \in B : \|q - p\|_\infty \le 10°\}$; **if** empty: **return** continuity (or IK) failure
take the $q \in B'$ nearest g ▷ B' usually has one element
end for
if the step from p moves some joint more than 0.5° **then**
re-walk it at 2 mm / 0.002 rad, nearest branch each step; keep where it arrives ▷ FineWalk
end if

15: $p \leftarrow$ the solved configuration; append it
16: **end for**
17: COLLIDES on the solved waypoints, middle first; **return** the waypoints or the failure and where
18: **end function**

The roadmap's configurations are only seeds and a yardstick: the first solve is seeded from the start node, every later one from the waypoint solved just before it (as the BiRRT extension does), and nothing requires the chain to reproduce a stored node. The joint line's configuration g only breaks ties: near a straight elbow or a straight wrist two branches can both lie within the gate of the previous waypoint, and then the one on the roadmap's side is taken. The waypoints hold the bar to 10^{-7} m, because each one is an IK solution of both arms at one bar pose.

Why (nearly) every step is walked again finely. The gate cannot tell a large honest step from a hop between branches: near a straight elbow or wrist the two branches can be closer than 10°. The production check found this in four rounds:

1. With 1 cm steps alone, 31 of 38 exact paths were walked by the production repair to the goal pose but arrived on another branch. On the best path (start 117) the cause was one 9.6° step at a 4.8° elbow where the chain had hopped to the mirror branch; walked at 2 mm the arm stays on its branch and arrives 9.4° away.
2. Re-walking every step above 2° at the production's resolution (FINEWALK: bar frame interpolated at 2 mm and 0.002 rad, warm IK nearest the previous solution, the gate at every step) raised the pass rate to 217 of 244 (89 %). All 27 failures were collisions, and none was a collision of our own waypoints or of the joint lines between them: the production walk had left our path at a smaller hop (4° at a 2° elbow, 8° at a 4° elbow) on an edge that had been tracked from its far end, and only collided further on.
3. Near such a point, which branch a walk keeps depends on the direction it walks in, so waypoints walked from one end and read backwards are not what a walk from the other end gives. Re-walking every step above 0.5° finely and tracking each edge in both directions (each direction keeping its own waypoints; a route uses an edge only in a direction that worked) left 25 failures in 268, again all collisions.
4. The remaining hops sat at nodes: an edge's chain had arrived within the gate of the shared node but on the mirror elbow solution (−4.3° against the node's +4.1°), and the next edge started from the node's own solution. Now an edge counts as tracked only when its chain ends on its node's own solution (next paragraph).

FINEWALK uses IK only (two solves per 2 mm step), so it roughly doubles the cost of tracking; collision is still checked at the 1 cm waypoints.

When an edge counts as tracked. A path is one chain, so where two edges meet, the next edge's chain must start where the arriving one ended. The next edge was tracked from the shared node, so an edge counts as tracked when its chain, started at one node, ends on the other node's own IK solution (both are solves at that node's bar pose, so they agree to 10^{-9} when they are the same solution). "Within the 10° gate" is not enough: near a straight elbow the mirror solution can be closer than that, and switching between two solutions at one pose is a branch flip no walk makes. (A seeded chain that continues a route from wherever it is need not end on the node; the route's next edge is then continued from there in the same way.) Usually an edge's chain does end on its node. It does not when the edge's joint line passes through a fully stretched elbow, where the two elbow branches meet: the corrected path, halfway between what the two arms imply, stays short of that point, so the chain cannot cross over and arrives on the mirror branch, 25–35° from the node on the elbow joint. This happened to 4 of 60 random B18 edges; neither finer steps nor the tie-break changed it. Such a direction is marked as *changing branch*; the edge stays usable in the other direction if that one arrives.

6.2 Exact gaps: tracking routes until they all track

Algorithm 11: ExactGaps — `refine.Tracker.exact_gaps`

```
function ExactGaps(query graph, starts within 30 mm)
    every collision-usable edge is untracked and assumed trackable
    repeat
        repeat
            (gap, parent) ← MinimaxToGoals over the edges not known to fail
            U ← untracked edges on each start's best route; TrackEdge them in parallel; mark failures
        until U = ∅
        assemble each start's route into one chain                          ▷ below
        where a chain cannot continue: drop the edge that left the roadmap's branch
    until every route assembles
    return each start's exact gap (its route's gap) and its exact path
end function
```

This is VerifyGaps with "track it" in place of "collision-check it", and exact for the same reason: tracking can only remove edges, so a route whose edges all track is the best route the graph has left. The *exact gap* of a start is therefore the smallest roadmap tolerance at which this roadmap yields a path that holds the bar exactly. *Assembly* joins a route's tracked edges into one chain and checks the gate at every node. Because every tracked edge arrives within the gate of both its nodes, this normally just works; where it does not (an edge tracked from its far end arrives a few degrees from the node), the next edge is tracked again, seeded from where the chain actually is, and if that fails the offending edge is dropped and the routes are searched again.

6.3 Tube draws around a route (way 2)

For a start whose exact gap is above the target (10 mm), new nodes are drawn around its current best route: a point is picked on the route's joint lines (points on edges over the target, or that failed to track, four times as often), the driving arm's joints are drawn from a Gaussian of 5° around it, and the other arm is solved as in Alg. 3. The new nodes join their 10 nearest neighbours within 30° (holding error measured; collision left to the tracker), and ExactGaps runs again on the enlarged graph. Up to three rounds, stopping after two rounds in a row with no start improving.

6.4 A BiRRT seeded with the route (way 1)

For a start the roadmap brings within 30 mm but that has no exact path even after the tube rounds (the 24 with the smallest roadmap gaps are tried), a task-space BiRRT bridges the failed edges of its original roadmap route with the production planner's own extension (`core.extend_toward`: warm IK, the 10° gate, the full collision check). Its two trees do not start at the two ends: tree A is the chain of exact waypoints from the start up to the first failed edge, tree B the chain from the goal back to the last failed edge. Random targets are bar poses in a tube around the route's bar poses (5 cm and 8.6° spreads; poses on failed edges four times as often), with a 10 % chance of aiming at the other tree's tip. A control run starts the trees from the two ends only, to separate the value of the exact chains from that of the tube.

6.5 The production check

Every stored exact path is finally re-solved by the production repair (`core.repair_joint_path`) at the benchmark's fine resolution (2 mm and 0.002 rad bar steps), with the production solver, gate and collision check. A path that passes is one the production planner would accept as is.

7 Running it in parallel

Sampling, edge checking and endpoint expansion are the same small computation thousands of times over, each independent. `WorkerPool` is a `ProcessPoolExecutor` with the `spawn` start method (the pattern of the benchmark's parallel library build); each worker opens its own headless PyBullet scene once, in its initializer, and keeps it. Processes, not threads: `pybullet_planning` keeps one global simulation and the collision check writes joints into it. Tasks and results are plain lists, so nothing that cannot cross a process boundary on Windows ever does; each edge task carries the configurations it needs (64 edges, about 12 kB). Results are re-sorted by chunk index, so the roadmap is the same for any worker count, and the parent never opens a scene: it does the nearest-neighbour search, the graph searches and the file writing. A worker that dies surfaces as a broken pool at once. Each worker sets `OMP_NUM_THREADS=1` so six or eight of them do not fight a running benchmark batch for cores; the default is six workers, eight were used for the runs below while no batch was running.

8 Storage and the dashboard

file	contents
`prm/index.json`	every roadmap run (only this tool writes it)
`prm/<run>/config.json`	settings, anchors, tallies per phase, timings, provenance (git state of every repository, versions, ssik artifacts)
`prm/<run>/roadmap.json.gz`	holding model; nodes as columns (configuration, both flange positions, their midpoint, bar pose, violation, kind, sampling mode, driver, anchor); edges as columns (ends, lengths, checked, free, where it collides, worst position and rotation error, where)
`prm/<run>/queries/<tag>.json.gz`	endpoint nodes and edges, the final verdict of every edge, every node's gap and parent, the paths with dense profiles, the per-start table
`prm/<run>/dashboard/`	the page (`prm.html`, also as `index.html`) with the shared scripts

The page reuses the benchmark dashboard's robot viewer, loaders and charts without changing them. Each node is a ball at the midpoint of the two flanges (grey sampled, purple anchors, green starts, blue goals; or coloured by gap). Balls are instanced meshes registered in the viewer's list of pickable clouds, so the existing click handler reports them. Clicking a ball poses the robot, lists the twelve joints, the node's role (anchor, which start or goal pair, which turn), its gap, and its edges sorted by holding error, each with a "walk" button. Edges are drawn green within the tolerance slider's value, amber up to three times it, red beyond. The best route of the clicked node is drawn and can be stepped through; at every step the page recomputes the holding error from the posed flanges, and a second, orange bar placed by the right flange shows the drift. Serve the data root and open `http://localhost:8000/prm/<run>/dashboard/`.

How to run it. From the repository root, with `HUSKY_BENCH_ROOT` set to the benchmark data root (PowerShell shown):

```
.venv\Scripts\python.exe scripts\prm_pilot.py build --name B18_g2_n4000 --library cdfm_v1 ‘
    --bar B18 --goals 2 --anchor-starts 40 --nodes 4000 --workers 8
.venv\Scripts\python.exe scripts\prm_pilot.py query --name B18_g2_n4000 --goal 2 ‘
    --starts all --tag all
.venv\Scripts\python.exe scripts\prm_pilot.py export --name B18_g2_n4000
.venv\Scripts\python.exe scripts\prm_pilot.py info --name B18_g2_n4000
python -m http.server --directory $env:HUSKY_BENCH_ROOT 8000
```

`-starts` also takes a class list (`free,blocked`) or start indices (`0,12`). Long runs should be started detached (e.g. `Start-Process` with the output sent to a log), not from a Claude Code background shell, which stops after ten minutes.

9 Core differences from the task-space BiRRT

P-1 Nodes are joint configurations that hold the bar.
The BiRRT's nodes are bar poses with an IK side cache (its D-1). Here a node *is* twelve joint values; the bar pose is derived from it. It is made by driving one arm and solving the other, so it holds the bar exactly.

P-2 No continuity gate.
Edges are straight joint-space lines cut at τ, so continuity holds by construction. The gate that stops 75 % of the BiRRT's extensions (D-6b) does not exist.

P-3 The holding constraint is soft along edges.
It is measured at every 2.5° and becomes the edge's score; the search either filters by a tolerance or pays a penalty above it.

P-4 Distance is the largest joint move.
Nearest neighbours use the raw joint ∞-norm, not the bar-corner feature metric (D-2): what matters for an edge is how far the joints move.

P-5 Sets, not points.
The start and goal are sets of every collision-free branch pair on two turns. The goal/home branch pairing heuristic and the requirement that both endpoints be fixed in advance (D-9) disappear; the search chooses.

P-6 One graph, many questions.
The graph keeps every alternative and is searched again under any tolerance or weight; there are no restarts (D-8) and no trees thrown away.

P-7 Collision checked where it decides the answer.
Endpoint edges are checked lazily, and the loops that do so are exact.

P-8 IK only at nodes.
One analytical solve per draw and 64 pairs per endpoint pose; none along any edge. The BiRRT solves both arms at every waypoint of every extension.

P-9 Parallel by design.
Sampling, edges and endpoints are split across worker processes, each with its own scene.

P-10 The answer is a number.
Every start gets a gap (the smallest tolerance at which it connects), so "near-feasible" and "disconnected" can be told apart.

10 Parameters and their defaults

parameter (flag)	default	meaning
`-nodes`	2000	sampled nodes kept (anchors come on top)
`-goals`, `-anchor-starts`	all, 40	goals whose pairs are anchors; library starts per goal used as anchors
`-anchor-turns`	both	anchors as given and on the home turn
`-mode-mix`	0.2/0.3/0.5	`anchor_box` / `anchor_gauss` / `anchor_line`
`-box-margin-deg`, `-gauss-sigma-deg`, `-line-sigma-deg`	90, 30, 10	spreads of the three draws
`-max-branches`	1	other-arm branches kept per draw, nearest the centre
`-k`, `-radius-deg`	10, 90	neighbours per node; longest edge (largest joint move)
`-collision-step-deg`	10	collision checks along an edge ($= \tau$)
`-viol-step-deg`	2.5	holding-error points along an edge
`-rot-deg-per-mm`	0.4	ρ in Eq. 2
`-k-endpoint`, `-radius-endpoint-deg`	10, 90	endpoint edges
`-tolerances-mm`, `-w-hold`	2,5,10,30; 10	query tolerances; penalty weight
`-endpoint-check`	lazy	collision of endpoint edges
`-workers`	6	worker processes
`-seed`	0	random seed

11 What the pilot measured

11.1 What was run

All runs used the benchmark's library `cdfm_v1` and its trimmed scenes (the scene key and git state are in each run's provenance record), the calibrated ssik solvers, eight worker processes on the 16-thread workstation (no benchmark batch was running), and the defaults of Section 10 unless stated.

- **B18, goal 2** — the goal whose 807 surveyed starts the BiRRT was run on in full (experiment 2, batch `20260929_1756`: two seeds per start, 21 of 807 starts solved at least once). Four roadmaps, with node budgets of 500, 1000, 2000 and 4000 sampled nodes, all anchored on goal 2 with 40 anchor starts (81 anchors in all: home, the 7 goal pairs, and 73 start configurations). Each roadmap was asked two separate questions, i.e. two queries on the same roadmap: one whose start set is only starts 0 and 12 (night 1: start 0 solved by 0 of 10 seeds, start 12 by 6 of 10), and one whose start set is all 807 starts.
- **B18, 2000 nodes, variations** — seeds 1 and 2, every IK branch kept per draw, and no start–goal line draws.
- **B3, goals 4, 11 and 16** — one 2000-node roadmap anchored on all three goals, queried per goal with the three night-1 starts.

Run folders are `<root>/prm/B18_g2_n<budget>/`, `B18_g2_n2000_<variation>/` and `B3_g4-11-16_n2000/`; the logs are in `<root>/prm/logs/`.

11.2 How fast a joint-space line leaves the holding surface

Figure 1 is the single most important measurement of the pilot, because it fixes how dense a roadmap has to be. Both ends of every edge hold the bar exactly, yet the straight line between them does not: the median worst error is 7.3 mm for edges shorter than 20° and 125.2 mm for edges of 60° and more. For the short edges that matter at small tolerances, sort the free edges by length (largest joint move) into three narrow bins and take the median worst holding error in each:

edge length (largest joint move, deg)	2.5–7.5	7.5–12.5	12.5–17.5
free edges in the bin (count)	51	642	2683
median worst holding error (mm)	1.0	3.6	6.9

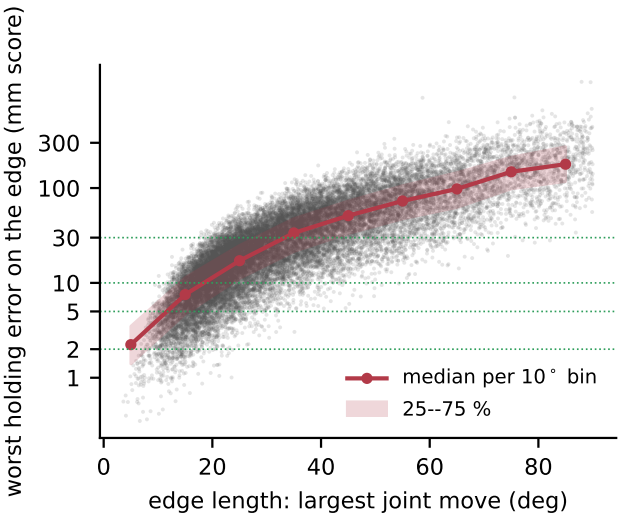


Figure 1: Worst holding error (score, Eq. 2) of every free edge of the 4000-node B18 roadmap against the edge's largest joint move. Dotted lines: the four query tolerances.

In words: an edge of about 5° drifts about 1 mm from the holding surface at its worst point; an edge as long as the continuity step $\tau = 10°$ drifts about 3.6 mm. Over this range the error grows roughly in proportion to the length rather than with its square, so the roadmap's edges are far from the regime where a straight chord hugs the curved holding surface. Two consequences follow. First, a sub-2 mm path needs a chain of edges of about 5° each; a roadmap sampled freely in twelve dimensions does not supply that density, so the gaps below are best read as *upper bounds* on how close a start is to the goal set. Second, the holding error, not collision, is what limits a joint-space roadmap: 93 % of candidate edges are collision-free, but only 1.1 % of those free edges stay within 2 mm, 7.5 % within 5 mm, 21 % within 10 mm and 57 % within 30 mm.

11.3 The roadmaps and what they cost

Table 1: The four B18 roadmaps (goal-2 anchors, seed 0, 8 workers), one column per node budget. A draw is one draw of the driving arm's six joints, followed by one IK call for the other arm. A component is a set of nodes joined by collision-free edges; the anchor rows say how many of the 7 goal anchors and the 73 start anchors sit in the largest one.

node budget	500	1000	2000	4000
sampled nodes kept: the node budget (count)	500	1000	2000	4000
all nodes, anchors included (count)	581	1081	2081	4081
driver-arm draws, one IK call each (count)	2656	5344	10688	21408
collision checks, sampling and edges together (count)	11818	23969	47606	89047
candidate edges measured (count)	2922	6094	13028	26908
collision-free edges (count)	2657	5586	12000	25056
components: separate islands of the free graph (count)	71	90	125	173
nodes in the largest component (count)	262	525	1285	2557
goal anchors in the largest component	3 of 7	3 of 7	7 of 7	7 of 7
start anchors in the largest component	24 of 73	25 of 73	27 of 73	27 of 73
sampling time (s, wall clock)	12	16	24	40
edge measuring time (s, wall clock)	28	74	103	181
whole build (s, wall clock)	51	100	138	232
edge phase speed-up (worker-seconds per wall-second)	7.4	7.8	7.9	8.0

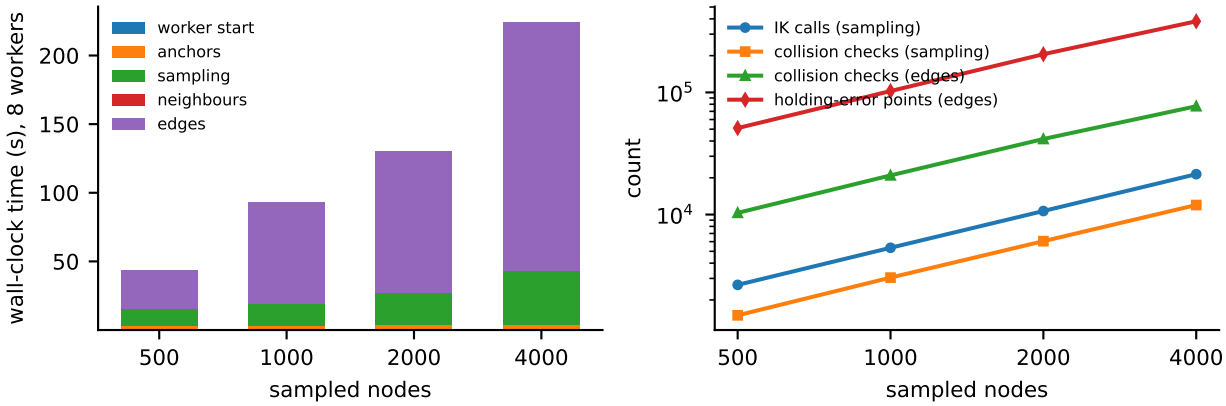


Figure 2: Left: where the build time goes. Right: the work counted, against the node budget.

Counting queries, as the meeting asked. Every build cost one analytical IK call per draw, and the sampler kept about one node in four and a half draws: of the 21 408 draws of the 4000-node build, 56 % were reachable by the other arm, and 39 % of those were collision-free. The line mode was by far the most productive (31 % of its draws kept, against 17 % for the Gaussian and 5 % for the box), which is why it carries half the mix. The edge phase then dominates: 77 076 collision checks for 26 908 edges, of which 1852 (6.9 %) collide. In worker time, collision checking was most of the whole build (93 %: 1571 of 1680 worker-seconds at 4000 nodes, 15.7–17.9 ms per check) and IK 2.7 % (2.1 ms per call); the 382 241 holding-error points cost nothing measurable. This inverts the pilot plan's expectation that IK would be the expensive operation: here it is the full-cell compas_fab check, exactly as in the BiRRT's own profile. The GPU question is therefore a question about batched collision checking, not about IK.

Parallel speed-up. Edge checking ran at 7.4–8.0 worker-seconds per wall-second on eight workers (Table 1), i.e. essentially linearly; sampling reached 6.0 at 4000 nodes and less on small budgets, where a round is only a few chunks. The query's endpoint expansion (38 384 branch pairs collision-checked for the 807 starts) ran at 7.2 and the endpoint edges at 7.7. The whole 4000-node build took 232 s of wall time.

What an "all-starts" query is, and why it takes 200 s. An all-starts query takes all 807 library start poses of the goal as its start set and gives each of them an exact gap. It is a survey of the start poses, not one planning query: on the same roadmap, the query with only starts 0 and 12 took 8.9 s. The all-starts query took 208–229 s whatever the roadmap size; for the 4000-node roadmap the 212 s split as follows.

phase	time (s)	what it depends on
solve and collision-check the start poses (38 384 branch pairs)	73	only the start poses and the scene: can be done once and stored
holding error of the endpoint edges (99 161 edges)	28	start poses and roadmap; not the goal
lazy gap verification (80 rounds)	104	the goal; about half is the 0.64 s spanning-tree search repeated every round, the rest is collision-checking 5461 edges (1231 blocked)
paths per tolerance	3	the goal

So a night of precomputation can remove nearly all of it: with the start poses and their edges solved and collision-checked once per roadmap, the verification loop ends after its first round. Section 11.12 builds and times exactly this.

Lazy checking is exact in practice too. On the smoke roadmap (bar B3, 333 nodes, 380 starts) the same query was run with endpoint edges checked lazily and eagerly: the gap of every one of the 380 starts (its smallest gap over all its branch pairs and turns) is the same both ways, and the lazy query took 151 s against 593 s (two workers).

11.4 Starts 0 and 12, and all 807 starts, against the node budget

Table 2: What each B18 roadmap answers, one column per node budget. The upper rows come from the query whose start set is starts 0 and 12 only; the lower rows from the query over all 807 starts. A gap is in mm of holding error (rotation folded in at 0.4° per mm); ∞: no route of free edges to any goal node.

node budget	500	1000	2000	4000
start 0: gap, best over its branch pairs and turns (mm)	82.6	41.8	41.4	21.3
start 0: gap of its library configuration alone (mm)	∞	∞	∞	∞
start 12: gap, best over its branch pairs and turns (mm)	7.9	5.7	5.1	4.5
start 12: gap of its library configuration alone (mm)	22.1	29.2	29.2	24.2
all 807 starts: smallest gap (mm)	7.9	5.7	4.2	2.9
starts that reach the goal set at all (count of 807)	650	657	657	657
median gap of the starts that reach it (mm)	49.4	41.8	30.5	24.3
starts with a gap of at most 2 mm (count)	0	0	0	0
starts with a gap of at most 5 mm (count)	0	0	1	11
starts with a gap of at most 10 mm (count)	13	17	24	36
starts with a gap of at most 30 mm (count)	114	115	171	462
all-starts query time (s, wall clock)	208	208	229	212

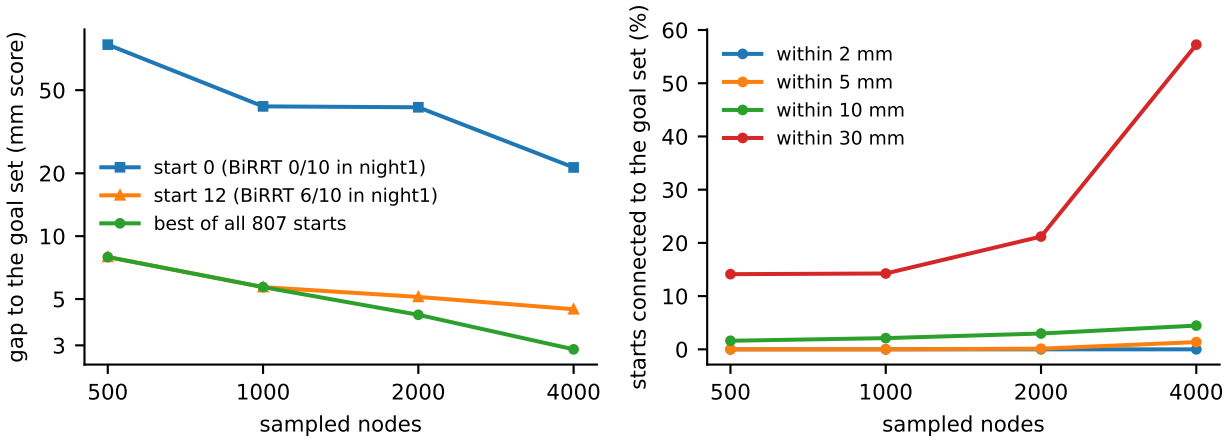


Figure 3: Left: gap to the goal set against the node budget for start 0, start 12 and the best of all starts. Right: share of the 807 starts that reach the goal set within each tolerance.

Three things stand out in Table 2 and Fig. 3.

Start 12 is near-feasible, start 0 is not. Start 12, which the BiRRT solved with 6 of 10 seeds in night 1, reaches the goal set within 7.9 mm on the smallest roadmap and within 4.5 mm on the largest. Start 0, solved by none of 10 seeds, sits at 82.6 mm on 500 nodes and still 21.3 mm on 4000 nodes. The reason is local: on the 2000-node roadmap, the 11–20 sampled nodes drawn around start 0's two anchors ended up a median 96–121° away from it (their other arm lands on a distant IK

branch, or they collide), and only two nodes lie within 30° of it; around start 12 the same numbers are 71–73° and 13–28 nodes. Start 0's gap on that roadmap is set by three long edges within the first five steps out of it (45–52°, 31–41 mm), not by anything near the goal.

The library's own start configuration is often the wrong one. For both starts the gap of the library configuration alone (the rows "gap of its library configuration alone" in Table 2) is much larger than the best over all branch pairs and turns; for start 0 it is unreachable at every budget. Over all starts, a different branch pair or turn beat the library configuration for 610 of the 657 reachable starts, and the best node sat on the home turn for most of them (Table 3). This is the quantitative form of the meeting's worry about start/goal branch pairing: fixing the start configuration before searching throws away most of the useful connections.

More nodes help, slowly. The best gap over all starts falls from 7.9 to 2.9 mm from 500 to 4000 nodes, the median over starts from 49 to 24 mm, and the number of starts within 30 mm from 114 to 462 (the least stable of these numbers: Table 6 shows it ranging from 171 to 405 across seeds at 2000 nodes). No start reaches the goal set within 2 mm at any budget. Exactly the same 150 starts are unreachable at 1000, 2000 and 4000 nodes: their start nodes do have edges, but into clusters of the roadmap that no edge of at most 90° joins to the goal's component (the 4000-node roadmap has 173 components); 136 of them are "broken" starts.

Table 3: The turn of each reachable start's best node (4000-node roadmap).

turn of the best node	starts
given	76
home	496
given and home	85

11.5 Which starts are near the goal: classes, carry anchors, and the BiRRT

Table 4: All 807 starts of B18 goal 2 on the 4000-node roadmap, by the library's straight-line class and by the BiRRT's outcome in experiment 2. "reach": starts with any route to the goal set; the last four columns count starts within each tolerance (mm).

group	starts	reach	median gap	≤2	≤5	≤10	≤30
free	28	28	6.8	0	7	19	28
blocked	43	29	11.5	0	4	11	26
broken	736	600	24.4	0	0	6	408
all	807	657	24.3	0	11	36	462
BiRRT solved (exp. 2)	21	21	6.8	0	8	15	21
BiRRT never solved	786	636	24.3	0	3	21	441

Table 5: The same starts by carry anchor. The last column counts the starts of each anchor that the BiRRT solved at least once in experiment 2.

carry anchor	starts	reach	median gap	≤10	≤30	BiRRT solved
back	306	195	23.4	36	171	21
horizontal	458	419	24.3	0	291	0
vertical	43	43	57.0	0	0	0

The roadmap ranks the starts the way the BiRRT did. The roadmap never ran the BiRRT and knows nothing about its results, yet the starts the BiRRT solved at least once in experiment 2 are the ones the roadmap puts closest to the goal set: all 21 of them are reachable, their median gap is 6.8 mm against 24.3 mm for the starts it never solved, 15 of them are within 10 mm and all

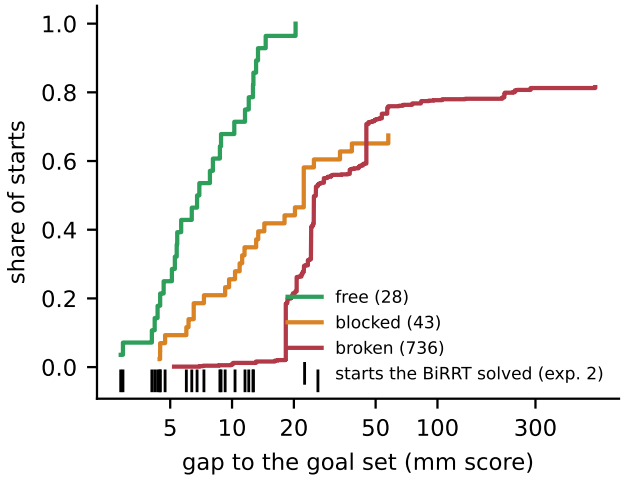


Figure 4: Share of starts whose gap is below a value, per library class (starts with no route are the missing share at the right). Ticks under the axis: the gaps of the starts the BiRRT solved in experiment 2.

21 within 30 mm (Table 4). The five starts with the smallest gaps were all solved by the BiRRT, and seven of the best ten. The library's straight-line classes are ordered the same way (medians 6.8, 11.5 and 24.4 mm for free, blocked and broken). The ranking is not perfect in the other direction: 21 of the 36 starts within 10 mm were never solved in experiment 2, but that batch ran only two seeds per start, and one of those 21 is start 12, which night 1 solved with 6 of 10 seeds.

The carry anchor matters most. Table 5 is the clearest single result. Every start the BiRRT solved is a "back" carry pose, and so is every start within 10 mm of the goal set; no horizontal or vertical start comes within 10 mm, and the vertical ones have a median gap of 57 mm. The benchmark's night-1 analysis found that production's start derivation always picks the horizontal anchor and succeeds 2.7 % of the time from it. For this goal the roadmap says why: from a horizontal carry pose no route of short joint-space steps reaches any of the goal's branch pairs.

11.6 The best path

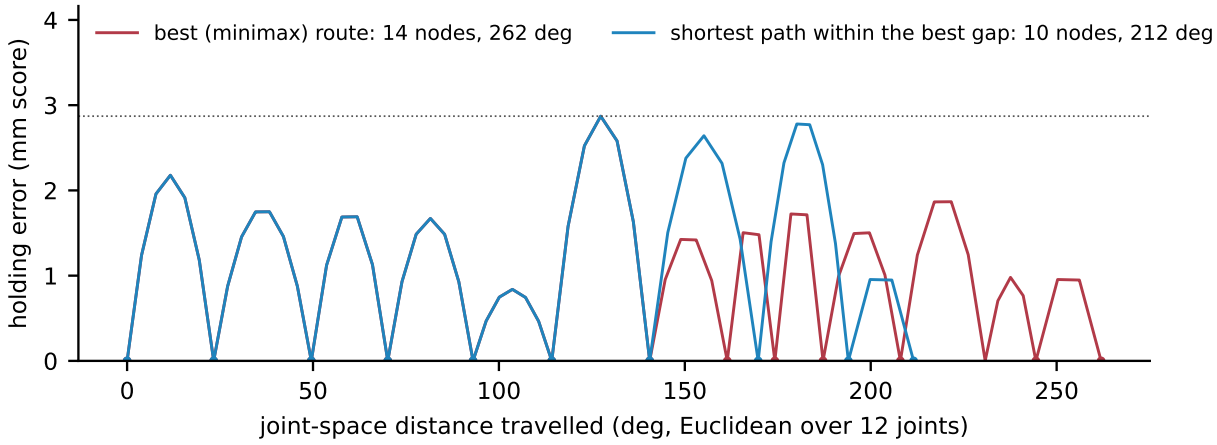


Figure 5: Holding error along the best route of the 4000-node roadmap (start 117) and along the shortest path within the same gap. Dots: roadmap nodes, where the error is zero. Dotted line: the gap.

The best route of the 4000-node roadmap starts at start 117 (a back, free-class start the BiRRT solved), on branch pair 38 and the home turn — not its library configuration — and ends at the goal's branch pair 7 (library numbering). It has 14 nodes and travels 4.57 rad in joint space, with no joint moving more than 2.47° between the dense points of the profile, so it contains no branch flip by construction. Its worst holding error is 2.87 mm. Figure 5 shows what that number is made of: the error rises from zero at each node to a peak in the middle of each edge and falls back to zero at the next node, and the peaks are all of similar height. Only two of the thirteen edges peak above 2 mm (at 2.2 and 2.9 mm); the others peak between 0.8 and 1.9 mm. There is no plateau and no single dominant spike: this is a resolution limit of the roadmap, not a sign that the start and the goal sit on disconnected sheets. Replacing those two edges by shorter ones — locally resampling around them, as the plan's refinement step proposes — is all that stands between this route and a sub-2 mm path. The filtered searches tell the same story from the other side: within 5 mm eleven starts connect, the shortest such path (7 nodes, from start 312, another BiRRT-solved start) has a worst error of 4.6 mm, and within 10 and 30 mm the shortest paths have 5–6 nodes.

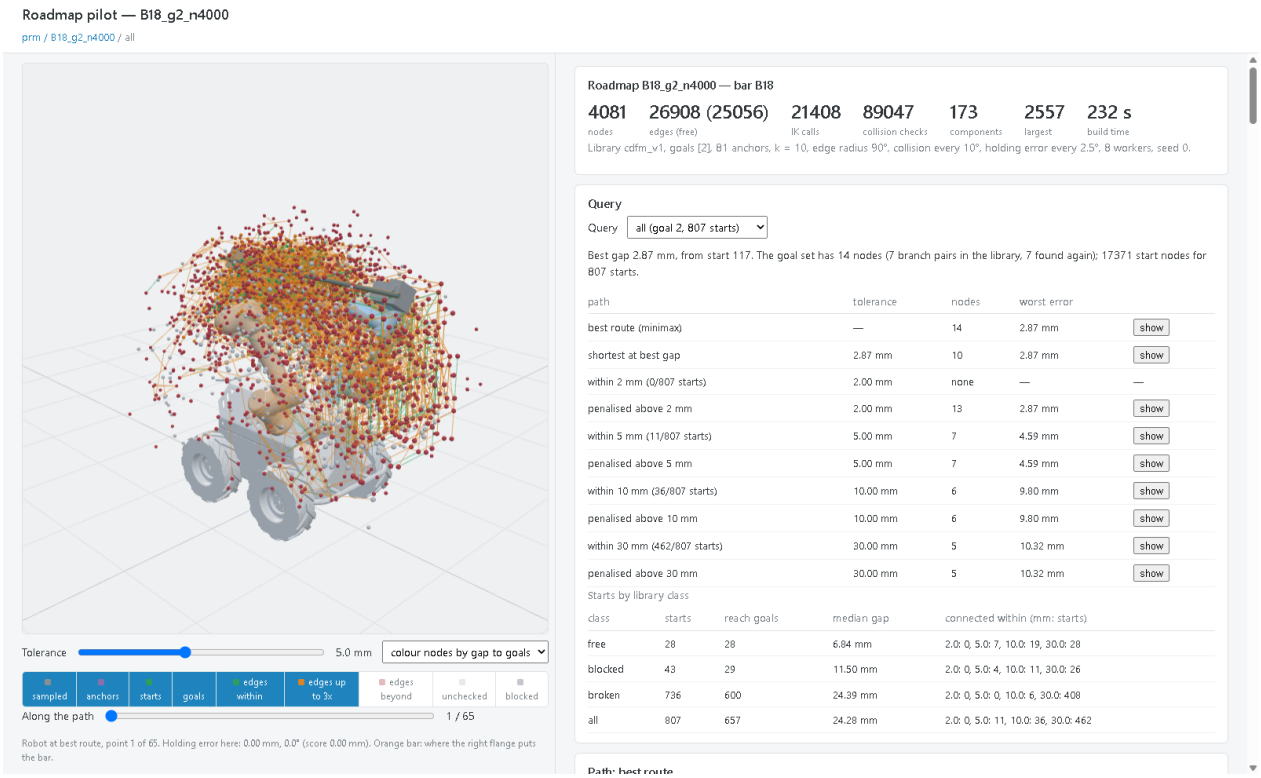


Figure 6: The roadmap page on the 4000-node B18 roadmap and the all-starts query: nodes at the midpoint of the two flanges, coloured by their gap to the goal set at the 5 mm slider setting, edges coloured by their worst holding error, the robot posed at the first point of the best route.

11.7 Seeds and two sampler choices

Seeds. Across three seeds at 2000 nodes the best gap is 3.8–5.0 mm, start 12's gap 3.9–5.1 mm, and 24–27 starts are within 10 mm: these numbers are stable. Start 0's gap (23.5–41.4 mm) and the count within 30 mm (171, 400, 405) are not; seed 0 happens to be the unlucky one for both, so the 30 mm counts of Table 2 should be read with a spread of that size.

Keeping every IK branch makes each draw yield more nodes, and that is exactly why it does worse at a fixed node budget. The budget counts nodes, not draws: 2000 nodes came from only 2272 draws (about 10 000 with $\beta = 1$), so four times fewer distinct bar poses were tried, and each was copied onto postures far from each other (flipped elbows and wrists, a median 182° from the

Table 6: The 2000-node B18 roadmap under three seeds and two sampler changes, one column per roadmap; rows as in Tables 1 and 2.

roadmap	seed 0 (default)	seed 1	seed 2	every IK branch kept	no line draws
sampled nodes kept: the node budget (count)	2000	2000	2000	2000	2000
all nodes, anchors included (count)	2081	2081	2081	2081	2081
driver-arm draws, one IK call each (count)	10688	9280	10304	2272	16768
collision checks, sampling and edges together (count)	47606	46198	46252	45350	58085
candidate edges measured (count)	13028	12983	12896	9500	11929
collision-free edges (count)	12000	11871	11909	8611	10174
components: separate islands of the free graph (count)	125	139	148	296	200
nodes in the largest component (count)	1285	1044	1034	755	1460
goal anchors in the largest component	7 of 7	3 of 7	3 of 7	3 of 7	3 of 7
start anchors in the largest component	27 of 73	24 of 73	24 of 73	23 of 73	48 of 73
sampling time (s, wall clock)	24	19	24	21	25
edge measuring time (s, wall clock)	103	82	98	78	97
whole build (s, wall clock)	138	114	133	110	132
edge phase speed-up (worker-seconds per wall-second)	7.9	7.9	7.9	7.8	7.9
start 0: gap, best over its branch pairs and turns (mm)	41.4	33.0	23.5	73.2	38.2
start 0: gap of its library configuration alone (mm)	∞	∞	∞	∞	∞
start 12: gap, best over its branch pairs and turns (mm)	5.1	3.9	5.0	5.3	13.7
start 12: gap of its library configuration alone (mm)	29.2	11.2	34.1	31.1	29.1
all 807 starts: smallest gap (mm)	4.2	3.8	5.0	5.2	13.7
starts that reach the goal set at all (count of 807)	657	651	658	658	650
median gap of the starts that reach it (mm)	30.5	24.3	24.5	44.0	25.5
starts with a gap of at most 2 mm (count)	0	0	0	0	0
starts with a gap of at most 5 mm (count)	1	4	3	0	0
starts with a gap of at most 10 mm (count)	24	25	27	16	0
starts with a gap of at most 30 mm (count)	171	400	405	112	406
all-starts query time (s, wall clock)	229	236	205	203	188

anchor on the smoke roadmap). Such copies find few neighbours within 90°, so the roadmap has *fewer* edges, not more (3500 fewer), falls into 296 components, and every quality number is worse (median gap 44 mm, 16 starts within 10 mm, 112 within 30 mm). The effort is spread over many branch sheets instead of filling the one or two that connect the starts to the goal. Section 11.10 gives it the same number of draws.

Without the start–goal line draws the roadmap still covers the space (406 starts within 30 mm, similar to the default seeds) but loses all precision: no start comes within 10 mm, and the best gap rises to 13.7 mm. This is not about the start itself. A start's gap is the worst edge anywhere on its best route to the goal, not something measured at the start; line draws put closely spaced nodes all along the corridor between starts and goal pairs, and without them the middle of every route is made of long edges, and a long edge drifts far from the holding surface (Fig. 1). The near-feasible routes of Fig. 5 are made almost entirely of line-draw nodes; they are what the roadmap's precision rests on. Line draws are also the most productive per draw (31 % kept, against 17 % for the Gaussian and 5 % for the box, at 4000 nodes), which is why the no-line roadmap needed 16 768 draws for the same 2000 nodes.

11.8 From roadmap routes to exact paths (E2)

Table 7: Refinement of the all-starts queries (Section 6), one column per roadmap or tracker setting. "Exact path from a route within t" means: some route all of whose roadmap edges score at most t was turned into waypoints that hold the bar exactly, are collision-free and never move a joint more than 10°.

refinement	4000 nodes knots 2.5°	4000 nodes end nodes only	2000 nodes seed 0	2000 nodes seed 1	2000 nodes seed 2
starts within 30 mm on the roadmap: the ones refined (count)	462	462	171	400	405
edges tracked (count)	3329	3388	1091	2373	2735
of which the tracker turned into exact waypoints	2737	2762	885	1961	2304
failed: no IK solution at some waypoint	65	45	14	35	37
failed: a joint jumped more than 10°	136	119	25	98	104
failed: a corrected waypoint collides	19	47	1	16	21
failed: changes elbow branch, so it cannot meet the next edge	372	415	166	263	269
edges re-tracked from a route's last waypoint at a node (count)	0	0	0	0	0
edges dropped because a route could not continue (count)	1	2	0	0	0
starts with an exact path from a route within 2 mm, of those the roadmap brings within 2 mm	0 of 0	0 of 0	0 of 0	0 of 0	0 of 0
starts with an exact path from a route within 5 mm, of those the roadmap brings within 5 mm	11 of 11	11 of 11	1 of 1	4 of 4	3 of 3
starts with an exact path from a route within 10 mm, of those the roadmap brings within 10 mm	34 of 36	33 of 36	24 of 24	25 of 25	27 of 27
starts with an exact path from a route within 30 mm, of those the roadmap brings within 30 mm	245 of 462	169 of 462	81 of 171	81 of 400	129 of 405
best start: smallest exact gap (mm)	2.87	2.87	4.20	3.78	4.97
its start index	117	117	227	313	12
waypoints of its exact path (count)	178	139	207	125	164
largest joint step between waypoints (deg)	2.76	3.55	2.69	2.57	2.31
holding error of its waypoints (mm)	4e-09	4e-09	4e-09	4e-09	4e-09
refinement time (s, wall clock)	1210	1085	483	858	902
stored exact paths (one per start, plus the best and the shortest ones) passing the production repair at 2 mm steps	249 of 249	–	–	–	–

Within 10 mm, almost every route becomes an exact path. Of the 36 starts the 4000-node roadmap brings within 10 mm of the goal set, 34 have an exact path from a route within 10 mm; within 5 mm, all 11 do; within 30 mm, 245 of 462. The whole refinement of the 807-start query took 1210 s on eight workers (57 tracking rounds, 3329 edges tracked). The best start is the same as the roadmap's (start 117): its exact path has 178 waypoints, holds the bar to 10^{-7} m, and moves no joint more than 2.8° between waypoints.

Why 10 mm is a good working tolerance. Fig. 7 shows that an edge's chance of tracking stays near 90 % up to 10 mm (89 % under 5 mm, 92 % between 5 and 10 mm; the failures there are mostly edges whose joint line passes a stretched elbow, which no tolerance helps) and then falls: 86 % between 10 and 15 mm, 79 % between 15 and 20 mm, 52 % between 20 and 30 mm. The joint correction grows at about 0.1° per mm of roadmap error (median over the tracked edges). So up to 10 mm the roadmap's routes are nearly as good as exact ones, and a 10 mm roadmap is the right thing to search before refining, as proposed. Of the 3329 tracked edges, 372 changed elbow branch, 136 hit the continuity gate, 65 met a pose one arm cannot reach, and 19 collided once corrected.

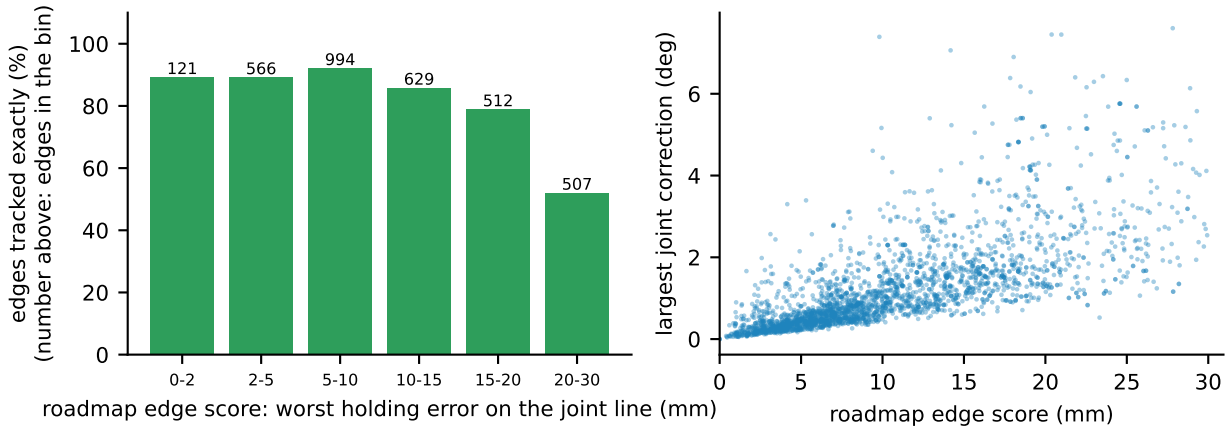


Figure 7: Every edge the refinement of the 4000-node roadmap tracked. Left: share of edges turned into exact waypoints, by the edge's roadmap score. Right: how far the correction moved the joints away from the straight joint line.

The same on other roadmaps, and why the knots matter. On the three 2000-node roadmaps (seeds 0, 1, 2), every start within 10 mm got an exact path (24 of 24, 25 of 25, 27 of 27); within 30 mm, 81 of 171, 81 of 400 and 129 of 405 (Table 7). So "within 10 mm on the roadmap" reliably means "an exact path exists", while within 30 mm it is roughly a coin toss that depends on the roadmap. Tracking a straight bar move between the two end nodes instead of following the bar along the joint line ("end nodes only") gives nearly the same answer within 10 mm (33 of 36) but clearly fewer exact paths within 30 mm (169 against 245): on longer edges the bar's straight move strays further from the joint line, and more of those edges change branch (415 against 372) or collide (47 against 19).

Exact paths from starts the BiRRT never solves. 164 horizontal starts have an exact path (from routes between 10 and 30 mm), although the BiRRT solved none of the 458 horizontal starts in experiment 2; 81 back starts do too, 245 in all, against the BiRRT's 21 (two seeds per start). 20 of the 21 starts the BiRRT solved are among them. This qualifies the carry-anchor finding above: from a horizontal carry pose no route stays within 10 mm, but many routes within 30 mm can still be made exact.

Table 8: The production check of every stored exact path of the 4000-node refinement: the production repair (`core.repair_joint_path`) re-solves each path at 2 mm and 0.002 rad bar steps with the production solver, gate and collision check.

carry anchor	back	horizontal	vertical	all
starts within 30 mm on the roadmap (count)	171	291	0	462
starts with an exact path from the roadmap (count)	81	164	0	245
passing the production repair at 2 mm steps	81	164	0	245
failing it on a collision between the 1 cm waypoints	0	0	0	0
failing it otherwise	0	0	0	0
starts the BiRRT solved at least once in experiment 2 (count)	21	0	0	21
of which the roadmap also gives a production-checked path	20	0	0	20

The production check. 245 of the 245 exact paths pass the production repair unchanged (Table 8; 3607 s on eight workers), 164 of them from horizontal starts and 20 from starts the BiRRT also solved: every path the refinement returns is one the production planner would accept as is.

Earlier versions of the tracker passed 7 of 38, then 217 of 244, then 243 of 268 paths; the difference was hidden branch hops near straight elbows, which the production walk at 2 mm does not take (Section 6). Removing them also removes the starts whose only exact route relied on such a hop, which is why the counts in Table 7 are lower than in the first version (298 starts within 30 mm then, 245 now; within 10 mm the count stayed at 34).

11.9 Tightening routes: tube draws (way 2) and a seeded BiRRT (way 1) (E3)

Table 9: The 4000-node refinement without and with three rounds of tube draws (each round: 200 draws around the routes of the 20 starts with the largest exact gap above 10 mm).

refinement	without tube draws	with 3 tube rounds
starts with an exact path from a route within 2 mm (count)	0	20
starts with an exact path from a route within 5 mm (count)	11	86
starts with an exact path from a route within 10 mm (count)	34	120
starts with an exact path from a route within 30 mm (count)	245	245
nodes added by tube draws (count)	0	10099
tube rounds run (count)	0	3
target starts improved, summed over rounds (count)	–	60
refinement time (s, wall clock)	1210	1888
stored exact paths passing the production repair at 2 mm steps	249 of 249	125 of 125

Table 10: The seeded BiRRT on the 24 starts still without any exact path after the tube rounds (smallest roadmap gaps first; all have roadmap gaps under 30 mm).

seeded BiRRT	seeded with the exact chains	from the two ends only
unresolved starts tried (count)	24	24
a path found (count)	0	1
passing the production repair (count)	0	1
median time of a passing run (s)	–	15
median IK solves of a passing run (count)	–	806
of the starts tried, solved by the plain BiRRT in experiment 2 (count)	0	0
time limit per start (s)	120	120

Tube draws work: the needed tolerance drops several-fold. Drawing nodes in a narrow tube (5°) around a start's current route and searching again is the refinement proposed in the meeting, and it does what was hoped (Table 9): with three rounds of about 3400 nodes each, the starts with an exact path from a route within 10 mm went from 34 to 120 (42 of them horizontal, none before), within 5 mm from 11 to 86, and within 2 mm from none to 20. Every target start improved in every round. The best route now needs only 0.89 mm of roadmap tolerance (start 206, whose roadmap gap was 23.4 mm before the tubes). The tubes do not add starts beyond the 245 that already had an exact path within 30 mm: they tighten routes, they do not create them. They cost 12 minutes on top of the 20-minute refinement, and all 125 exact paths stored from the tube-refined routes pass the production repair as well.

The seeded BiRRT does not bridge what the tracker cannot. On the 24 starts left without any exact path (Table 10), the BiRRT seeded with the route's exact pieces found no path in 120 s each; started from the two ends only, with the same tube of samples, it found one (start 573, in 15 s,

and it passes the production check). Of the 49 failed edges on these 24 routes, 30 are branch changes at a stretched elbow (Section 6) and 17 pass through poses one arm cannot reach; a task-space tree grown from the exact pieces meets the same obstacles, so seeding it with them does not help (it may even hurt: the trees start deep inside the two branch sheets that do not meet). These are the hardest starts of the goal, so the sample says little about the BiRRT in general, but for this refinement the roadmap side (tubes) is clearly the better lever.

11.10 Wider edges (E1) and every IK branch at the same number of draws (E4)

Table 11: E1: the 4000-node roadmap rebuilt with edges of up to 160° (same seed, so the same sampled nodes; only the edges differ). E4: every IK branch kept, given about the same number of draws (10 848) as the default 2000-node roadmap (10 688 draws, one branch per draw), next to the earlier every-branch roadmap that stopped at 2000 nodes (2272 draws).

roadmap	4000 nodes 90° (ref.)	4000 nodes 160° (E1)	every branch 2272 draws	every branch 10k draws (E4)	one branch 10k draws (ref.)
sampled nodes kept: the node budget (count)	4000	4000	2000	9400	2000
all nodes, anchors included (count)	4081	4081	2081	9481	2081
driver-arm draws, one IK call each (count)	21408	21408	2272	10848	10688
collision checks, sampling and edges together (count)	89047	101487	45350	232212	47606
candidate edges measured (count)	26908	29089	9500	56748	13028
collision-free edges (count)	25056	25928	8611	51717	12000
components: separate islands of the free graph (count)	173	68	296	533	125
nodes in the largest component (count)	2557	3873	755	5839	1285
goal anchors in the largest component	7 of 7	7 of 7	3 of 7	3 of 7	7 of 7
start anchors in the largest component	27 of 73	67 of 73	23 of 73	49 of 73	27 of 73
sampling time (s, wall clock)	40	35	21	99	24
edge measuring time (s, wall clock)	181	226	78	658	103
whole build (s, wall clock)	232	272	110	769	138
edge phase speed-up (worker-seconds per wall-second)	8.0	6.0	7.8	6.0	7.9
start 0: gap, best over its branch pairs and turns (mm)	21.3	21.3	73.2	21.8	41.4
start 0: gap of its library configuration alone (mm)	∞	648.9	∞	∞	∞
start 12: gap, best over its branch pairs and turns (mm)	4.5	4.5	5.3	4.3	5.1
start 12: gap of its library configuration alone (mm)	24.2	24.2	31.1	24.2	29.2
all 807 starts: smallest gap (mm)	2.9	2.9	5.2	3.4	4.2
starts that reach the goal set at all (count of 807)	657	801	658	713	657
median gap of the starts that reach it (mm)	24.3	25.0	44.0	23.1	30.5
starts with a gap of at most 2 mm (count)	0	0	0	0	0
starts with a gap of at most 5 mm (count)	11	11	0	12	1
starts with a gap of at most 10 mm (count)	36	36	16	47	24
starts with a gap of at most 30 mm (count)	462	462	112	471	171
all-starts query time (s, wall clock)	212	553	203	340	229

E1: 160° edges connect more starts, but only through useless edges. Widening the edge limit from 90° to 160° merged the roadmap from 173 components into 68 and made 801 of the 807 starts reachable instead of 657. But the 144 newly reachable starts reach the goal set only through edges that break the holding constraint by 220 to 1787 mm (median 984 mm; none within 100 mm): a straight joint line of more than 90° simply does not hold the bar. Of the 657 starts reachable

both ways, 11 got a smaller gap, all of them above 30 mm before and after. Nothing within 30 mm changed (best gap 2.87 mm; 11, 36 and 462 starts within 5, 10 and 30 mm), while the all-starts query took 553 s instead of 212 s (272 verification rounds instead of 80). The 90° limit was not hiding useful connections; the islands it leaves apart are joined only by moves no path can use.

E4: every IK branch is worth it per draw, not per second. With the node budget counting nodes, keeping every branch looked worse (Table 6). Given the same number of draws as the one-branch roadmap, it is clearly better: 47 starts within 10 mm and 471 within 30 mm, against 24–27 and 171–405 for the three one-branch seeds, and a best gap of 3.4 mm against 3.8–5.0 mm. That is about what the one-branch roadmap reaches with twice the draws (4000 nodes: 36 and 462). But it has 9481 nodes, 56 748 candidate edges and 232 000 collision checks, and took 769 s to build against 232 s for the 4000-node one-branch roadmap; its graph is also more broken (533 components, only 3 of the 7 goal anchors in the largest). Per collision check the one-branch roadmap is still ahead: 36 starts within 10 mm for 89 000 checks, against 47 for 232 000. The other branches help because they are other routes to the same goal pairs, which is also the motivation for the start and goal *sets*.

11.11 Re-calibrating the three carry poses (E5)

The library's start poses come from three carry anchors that were never calibrated; the horizontal one, for example, puts the grasp midpoint at $x = 0.40$ m, 19 cm closer to the robot than the demo pose it was taken from. The search (`holding_prm/carry.py`, `scripts/prm_carry_search.py`) keeps the three classes and their bar axes (horizontal: along y; vertical: along z; back: along x) and varies the rest: the grasp midpoint on a 10 cm grid over a $0.6 \times 0.6 \times 0.6$ m box per class, and the roll about the bar in 30° steps (4116 candidates per class). A carry pose is stored as these three numbers, so the same pose applies to any bar (the bar's orientation follows from its grasps, as the planner's anchors do).

- *Stage 1*, on bars B18 and B3: the candidate must be holdable on both bars; then ranked by how many of its branch pairs are comfortable (at least 29° from every joint limit, and elbow and wrist at least 30° from straight), and by how many of 24 nudges (5 cm and 10 cm shifts, 15° and 30° turns, about each base axis) it survives. A pose close to the robot body loses its nudges toward the body. The 30 best of each class, spread over places and rolls, plus today's anchors, got the nudges.
- *Stage 2*: the 5 best of each class and today's three anchors were given to the roadmap as the start set, against 8 goals of B18 and 6 of B3, one 3000-node roadmap per goal (a first try with one 4000-node roadmap over 8 goals was too sparse: 311 components), each query refined to exact paths.

What the search says. The carry *class* matters more than the exact place. Back carry is the best class: today's back anchor and the best back candidate both reach 4 of the 14 goals with an exact path from a route within 10 mm; moving it does not help much (the best candidate has a smaller median exact gap, 5.6 mm). Horizontal carry is where today's anchor is clearly wrong: it reaches none of the 14 goals, while the best candidate, raised to $z = 1.2$ m and moved forward and left (midpoint $(0.45, 0.10, 1.20)$ m, roll 60°), reaches 2 within 10 mm and 4 with some exact path. No vertical pose works: the best reaches one goal at 20 mm. And for most goals no carry pose of any class reaches within 10 mm on a 3000-node roadmap, which says the goals, not the carry poses, are the hard part at this density. The current picks are in `asset/carry_poses.json`, marked provisional, with the three best of each class listed for a person to choose from; nothing in the planner or the library uses them yet.

11.12 Precomputing overnight (E6)

The night before, only the bar and its grasps are known; the goal (the base's parked pose relative to the structure) is known only on site. Everything that does not depend on the goal can therefore

Table 12: The carry search, one column per class. Midpoints in the mobile-base frame; "goals with an exact path" count, over the 14 goals of stage 2, those the carry pose reaches with a path that holds the bar exactly.

carry class	back	horizontal	vertical
grid candidates holdable on both bars (of 4116 per class)	1277	2569	370
today's anchor at its best roll: grasp midpoint (m), roll	(0.10, 0.00, 1.04), −60°	(0.40, -0.04, 0.86), −60°	(0.71, 0.00, 0.70), −60°
nudges survived on its worst bar	24 of 24	24 of 24	23 of 24
goals with an exact path from a route within 10 mm (of 14)	4	0	0
goals with any exact path (route within 30 mm)	6	0	0
best candidate: grasp midpoint (m), roll	(0.00, -0.10, 1.10), −30°	(0.45, 0.10, 1.20), 60°	(0.80, 0.00, 0.60), 0°
nudges survived on its worst bar	24 of 24	24 of 24	20 of 24
goals with an exact path from a route within 10 mm (of 14)	4	2	0
goals with any exact path (route within 30 mm)	5	4	1
median exact gap where it has one (mm)	5.6	11.6	19.9

move to the night: the roadmap itself, with the three carry poses' branch pairs as nodes and every edge collision-checked, and the exact tracking of every roadmap edge within 10 mm (`prm_pilot.py track`, which writes the verdicts and waypoints to `tracked_10mm.json.gz`). Online, a query only solves the goal pose and the three carry poses, measures their edges, and searches; the refinement then tracks only the few edges that touch the goal (`refine -pretracked`). The test uses goals the roadmap was *not* anchored on (B18 goals 2, 10, 20, 31, 47) and the provisional carry poses.

Online time is small; coverage is the problem. Without the 10 s of worker start-up, an online query for an unseen goal took about 6 s (most of it reading the roadmap file; solving the 4 poses took 1.5 s, the gaps 0.1–0.3 s) and the refinement about 2 s (Table 13). But none of the five unseen goals came within 10 mm of a carry pose: best gaps of 17–56 mm, and no route at all to goal 47. A 6000-node roadmap spread over 17 goals is far sparser per goal than the 4000 nodes the goal-2 roadmap spent on one goal, and only 1218 of its 33 582 free edges are within 10 mm. What the night has to buy is density, not speed: Table 14 repeats the test with a 30 000-node roadmap anchored on 45 goals.

A night-sized roadmap: closer, and the carry set becomes the limit. The 30 000-node roadmap took 35 minutes to build and 89 minutes to track its 16 883 edges within 10 mm (16 013 track), about two hours in all on eight workers; its largest component holds 320 of the 405 goal anchors and 85 of the 88 carry-pose anchors. The online cost stayed about the same (Table 14): about 16 s per query, of which 8 s start the workers, 2 s are the query's own phases and most of the rest is reading the five-times larger roadmap file – both would be paid once by a planner that keeps its workers and the roadmap loaded for the session. Unseen goal 10 now has an exact path (from a 9.8 mm route; it passes the production repair), goals 20 and 31 come within 12–13 mm, goal 2 stays at 40 mm and goal 47 has no route. Goal 2 is telling: the all-starts query of Section 11.8 found routes of 2.9 mm to it from the library's back starts, but from the three fixed carry poses the best is 40 mm. Fixing a single pose per class gives up most of the start-set freedom that made the roadmap work; a small family of carry poses per class (a few places and rolls), all in the night roadmap and all in the online start set, costs nothing online and is the obvious next step.

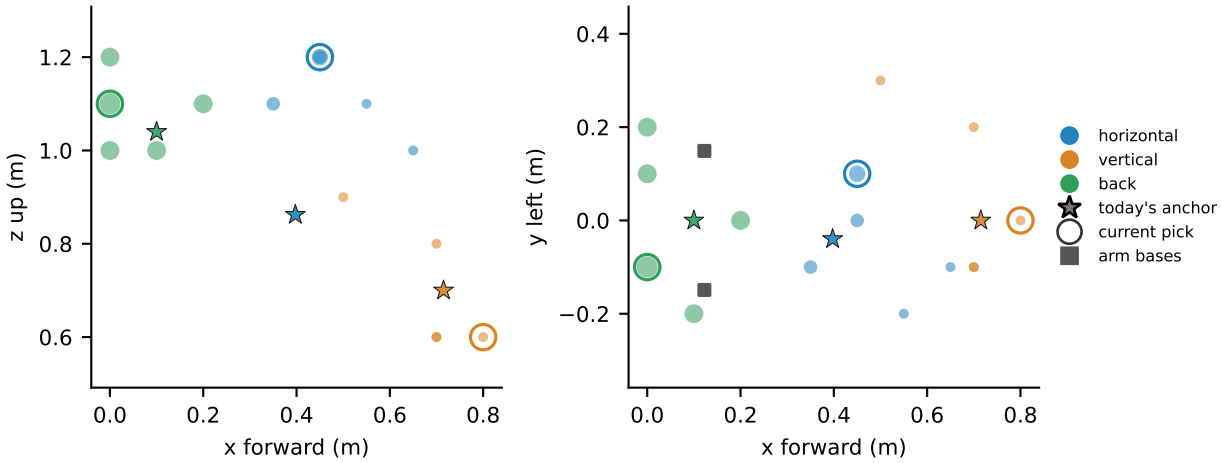


Figure 8: Stage-2 carry candidates by their grasp midpoint (side and top view; marker size grows with the number of goals reached within 10 mm). Stars: today's anchors; rings: the current picks.

11.13 Bar B3

On bar B3 the roadmap orders the three night-1 starts of each goal as the BiRRT did (Table 15). In goal 4, start 42 (8 of 10 seeds) has a 6.5 mm gap, start 13 (4 of 10) 16.2 mm, and start 0 (0 of 10) no route at all. In goal 16 the best-solved start (13, 8 of 10 seeds) has the smallest gap, 7.5 mm, against 17.1 mm for the other two. In goal 11, where starts 13 and 221 were each solved twice, start 13 has 7.6 mm and start 221 16.7 mm. Start 0 — production's horizontal pick in all three goals — has the largest gap or ties for it. Again the library configuration is rarely the best: start 13 of goal 11 reaches the goals within 7.6 mm on its best branch pair but only within 35.6 mm on its library configuration.

11.14 What this says, and what it does not

Observations.

(i) **The pilot's two questions have first answers.** (1) On B18 goal 2, a joint-continuous path exists whose holding error stays under 2.87 mm, from a start the BiRRT also solved, found with 4000 nodes: 23 024 IK calls (21 408 to build, 1616 to solve the 808 endpoint poses) and 141 501 collision checks (89 047 to build, 52 454 for the query over all 807 starts), in 232 s plus 212 s on eight workers. No path under 2 mm was found at any budget. (2) Where the roadmap fails, the failure is local and of known shape: the best route's error is a row of similar peaks at 1–3 mm, two edges over 2 mm; start 0's is a few long edges right next to the start.

(ii) **The gap is a useful difficulty measure without running the BiRRT.** It separates the starts the BiRRT solved from those it did not (median 6.8 against 24.3 mm), orders the library's classes, and on B3 orders the night-1 starts of three goals as the BiRRT did.

(iii) **The carry anchor, and the branch pair, are decided wrongly today.** For B18 goal 2 only back-carry starts come within 10 mm, while production picks the horizontal anchor; and for 610 of 657 reachable starts some other branch pair or turn beats the library's own start configuration. A roadmap query over a set of starts makes both choices by search.

(iv) **Global density is the wrong lever.** Doubling the node budget from 2000 to 4000 lowered the best gap from 4.2 to 2.9 mm and did not reach any start that was unreachable at 1000. The holding error of an edge is set by its length (Fig. 1), and random nodes do not shorten the edges of one particular route fast enough.

(v) **Collision checking is the cost.** 93 % of build worker time; IK is under 3 %. The work parallelises almost perfectly on the CPU (8.0x on 8 workers for edges). A GPU, if ever, would be for batched collision checks.

Table 13: A night roadmap of 6000 nodes anchored on 17 other goals of B18 and the three carry poses, then five online queries for unseen goals (8 workers). Worker start-up is listed apart: a planner that runs for a whole session pays it once.

unseen goal	2	10	20	31	47
night: nodes, of which anchors (count)	6248, 248				
night: goals the roadmap is anchored on (count)	17				
night: build time (s)	821				
night: collision-free edges, of which within 10 mm (count)	33582, 1218				
night: tracking the edges within 10 mm (s)	151				
night: components of the free graph (count)	318				
online query: start the workers (s)	9.8	10.0	9.6	10.0	10.0
solve the goal and the 3 carry poses (s)	1.5	1.4	1.5	1.4	1.4
holding error of their edges (s)	0.2	0.8	0.2	0.5	0.0
gaps, with lazy collision checks (s)	0.1	0.2	0.1	0.3	0.1
whole query (s)	16.5	16.4	15.4	15.9	15.2
online refinement with the night's verdicts (s)	12.0	12.1	11.6	11.7	12.1
of which starting its workers (s)	9.9	10.1	9.7	9.8	10.0
best gap from the carry poses to the goal (mm)	56.2	16.6	32.5	19.0	∞
carry poses with an exact path within 10 mm (of 3)	0	0	0	0	0

What the numbers do not say. A gap is an upper bound on the smallest holding error a path needs: a denser or better-placed roadmap can only lower it. "Unreachable" means unreachable in this roadmap within 90° edges, not infeasible (90° is the largest move of any single joint along one edge, in either direction, so a band of 180°; Section 11.10 widens it to 160°). Each budget was built once (seed 0); at 2000 nodes a different seed moves the best gap by about 1 mm and the 30 mm count by more than a factor of two (Table 6). Collision along an edge is checked every 10° of joint motion, coarser than the production planner's 2 mm pose steps, so a roadmap path has to be re-solved exactly (`core.repair_joint_path`, step 7 of the plan) before it can be executed; the pilot does not do this yet. And all of this is one goal of one bar, plus three goals of B3.

Second version: what the new experiments add.

(vi) **A 10 mm roadmap route is nearly an exact path.** Tracking the bar along a route and solving both arms at every step turned 34 of the 36 routes within 10 mm, and 245 of 462 within 30 mm, into exact paths; all 245 pass the production repair unchanged. That is about twelve times the 21 starts the BiRRT solved, including 164 horizontal starts it never solved (Section 11.8).

(vii) **The hard part of exactness is branches, not millimetres.** Most edges that fail to track have a joint line through a stretched elbow, where the exact path ends on the mirror branch; and four rounds of the production check were needed to find every way a coarse chain can hop between branches near such a point (Section 6).

(viii) **Wider edges do not help; more branches help per draw** (Section 11.10).

(ix) **Carry poses: the class matters more than the place.** Back carry is best and today's back anchor is about as good as any; today's horizontal anchor reaches none of 14 goals, a raised one reaches 2–4; no vertical pose works (Section 11.11).

(x) **Overnight precomputation makes the online part a few seconds**, but the night has to buy density: a 6000-node roadmap over 17 goals reached none of 5 unseen goals within 10 mm (Section 11.12).

Next steps suggested by the data.

Table 14: The same test with a night roadmap of 30 000 nodes anchored on the 45 other goals of B18.

unseen goal	2	10	20	31	47
night: nodes, of which anchors (count)	30494, 494				
night: goals the roadmap is anchored on (count)	45				
night: build time (s)	2128				
night: collision-free edges, of which within 10 mm (count)	190571, 16883				
night: tracking the edges within 10 mm (s)	5319				
night: components of the free graph (count)	682				
online query: start the workers (s)	7.7	7.9	7.7	7.5	7.7
solve the goal and the 3 carry poses (s)	1.3	1.1	1.4	1.2	1.2
holding error of their edges (s)	0.3	0.5	0.3	0.3	0.0
gaps, with lazy collision checks (s)	0.3	0.2	0.2	0.2	0.9
whole query (s)	17.4	16.5	16.4	16.5	15.7
online refinement with the night's verdicts (s)	13.7	16.1	12.4	12.5	12.1
of which starting its workers (s)	8.0	7.7	7.8	7.7	7.5
best gap from the carry poses to the goal (mm)	40.2	9.8	11.7	13.1	∞
carry poses with an exact path within 10 mm (of 3)	0	1	0	0	0

Table 15: Bar B3, one 2000-node roadmap anchored on goals 4, 11 and 16; the three night-1 starts of each goal. "night 1": seeds (of 10) the BiRRT solved.

goal	start	carry	class	night 1	gap	lib. gap	best pair
4	0	horizontal	broken	0/10	∞	∞	44
4	13	back	blocked	4/10	16.2	25.4	63
4	42	back	free	8/10	6.5	14.9	51
11	0	horizontal	blocked	0/10	16.7	16.7	44
11	13	back	free	2/10	7.6	35.6	46
11	221	horizontal	free	2/10	16.7	35.6	45
16	0	horizontal	broken	0/10	17.1	17.1	44
16	13	back	free	8/10	7.5	11.7	46
16	23	horizontal	free	3/10	17.1	17.1	47

1. **Bring the refinement into the planner**: a `use_prm` variant of `plan_constrained_dual_arm` that queries a stored roadmap with the carry poses as the start set, refines, and hands the production repair a path, and a benchmark variant so it can be batched against the BiRRT.
2. **Collision distance along the route**: the production check passes everything now, but only collisions at 1 cm waypoints are checked during tracking; a clearance cost would keep routes away from contact.
3. **Denser night roadmaps**: decide the node budget per goal region from the 30 000-node test, and whether goals of one structure can share it.
4. **Choose the carry poses** from the three listed per class, then re-run the night build with them.

12 Reading guide

Line numbers are from the tree as written on 30 September 2026.

read	at	this report
`violation_from_tools`, `holding_violation`	`holding.py:295`, `:320`	Alg. 1
`pose_gap`	`holding.py:130`	the angle
`interpolate_confs`, `rebranch12`	`holding.py:387`, `:421`	continuity; turns
`make_collision_fn`	`sampling.py:68`	collision
`arm_branches`	`sampling.py:148`	IK on the cached base
`draw_driver`, `lift_draw`, `sample_chunk`	`sampling.py:278`, `:323`, `:362`	Algs. 2, 3
`expand_bar_pose`	`sampling.py:413`	Alg. 6
`candidate_pairs`, `evaluate_edge`, `edge_cost`	`roadmap.py:55`, `:163`, `:209`	Alg. 4
`minimax_to_goals`, `lazy_search`	`search.py:169`, `:222`	Algs. 7, 8
`WorkerPool.run`, `init_worker`	`parallel.py:229`, `:56`	Section 7
`make_anchors`, `build_roadmap`	`pilot.py:289`, `:409`	Alg. 5
`query_roadmap`	`pilot.py:617`	Alg. 9